📊 Social Media Impact Analysis – Instagram Usage & Wellbeing

Project Overview

This project analyzes Instagram usage behavior and its relationship with user demographics, lifestyle patterns, and engagement metrics. The goal was to understand **how time spent, content consumption, and engagement vary across age groups, employment status, and health indicators**, and to translate raw social media data into actionable insights using SQL and Tableau.

Business Problem

Social media platforms and marketers need to understand:

- Which user segments are most engaged
- How usage patterns differ by age and lifestyle
- Whether excessive usage correlates with health and wellbeing indicators

This project simulates a **real-world analytics use case** where behavioral data is used to inform **product strategy, digital wellbeing initiatives, and content optimization**.

Dataset Description

- Source: Public Instagram usage dataset (CSV files)
- Records: User-level behavioral and demographic data
- Key tables/files used:
    - `insta_raw` (core dataset)

    - Time Spent metrics
    - High KPI indicators
    - Employment Status
    - Content Preferences by Age
    - Age Group & Health metrics

Tools & Technologies

- **SQL (PostgreSQL)** – Data cleaning, aggregation, KPI calculation
- **Tableau Public** – Interactive dashboards & visual storytelling
- **Excel** – Initial data validation and formatting
- **Calculated Fields (Tableau)** – Age grouping, engagement metrics

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Data Preparation & Modeling

- Cleaned and standardized raw CSV files
- Validated numeric fields (age, time spent, engagement metrics)
- Created derived fields such as:
    - Age Groups
    - Engagement intensity
    - Average daily activity metrics
- Used SQL to:

    - Aggregate user-level engagement
    - Calculate averages and counts by demographic segments
    - Identify high-engagement user cohorts

Key SQL Analysis Examples

Examples of analysis performed:

- Count of users by age group
- Average daily Instagram usage by employment status
- Engagement metrics (likes, comments, DMs) per user
- Identification of high-usage vs low-usage segments

(SQL was used as the analytical layer before visualization.)

Tableau Dashboard Design

Created an interactive dashboard with the following key visuals:

📌 Core Visualizations

- **User Distribution by Age Group**
- **Average Daily Time Spent on Instagram**
- **Engagement Metrics by Age Group**
- **Content Type Preferences Across Demographics**
- **Employment Status vs Usage Intensity**
- **Health Indicators vs Social Media Usage**

📌 Interactivity

- Filters by age group, employment status, and content preference
- Tooltips to display detailed engagement metrics
- Clean layout designed for non-technical stakeholders

Key Insights

- Younger age groups showed **higher engagement frequency** but shorter session lengths
- Employed users demonstrated **more consistent but time-bound usage**
- High engagement correlated with increased daily active minutes
- Content preferences varied significantly across age segments
- Excessive time spent showed potential correlation with stress indicators (observational, not causal)

Business Impact

This analysis demonstrates how social media usage data can be used to:

- Inform **content personalization strategies**
- Identify **high-risk overuse segments**
- Support **digital wellbeing monitoring**
- Enable data-driven decisions for marketing and platform optimization

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Skills Demonstrated

- Data cleaning and validation
- SQL-based analytical thinking
- KPI design and segmentation
- Dashboard storytelling for stakeholders
- Translating raw data into business insights

Deliverables

- SQL analysis scripts
- Interactive Tableau dashboard (Tableau Public)
- Documented analytical insights

Next Enhancements (Future Scope)

- Time-series analysis of usage trends
- Predictive modeling for engagement risk
- Cross-platform social media comparison
- Correlation analysis between health metrics and usage patterns
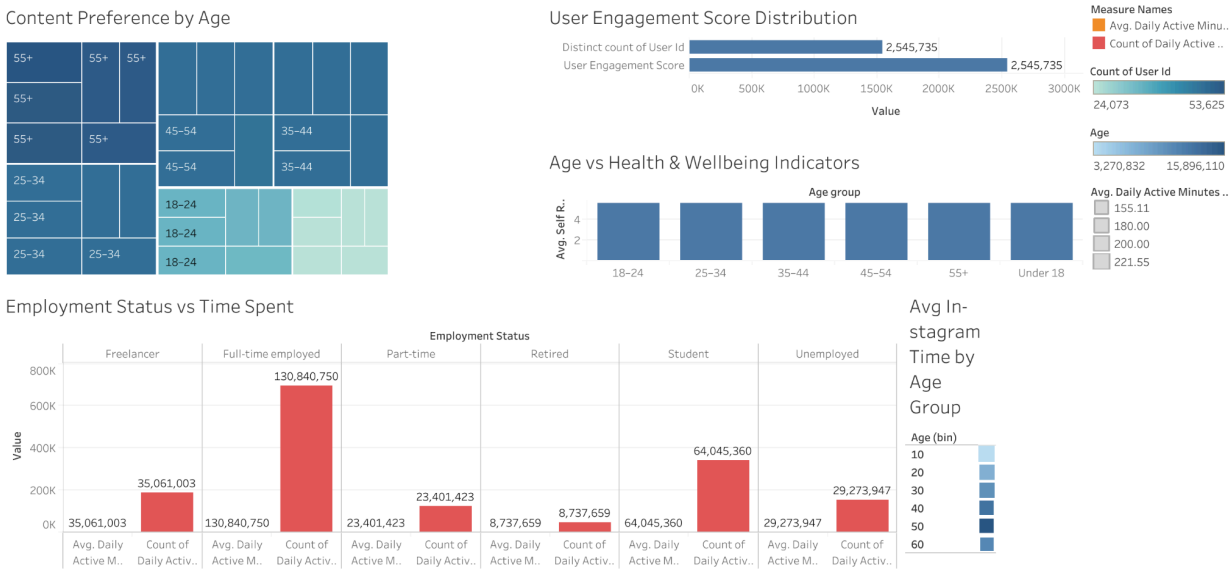
Tableau Dashboard